Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Development of a "Field-Ready" Hybrid AI Optimization Platform for Waste Plastic HTL: Integrating Real-time Vision Sensing and Global Knowledge Mining			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Knowledge Services		Energy & Resources	
	Medium Category	Small Category	Medium Category	Small Category
	R&D / Engineering Services	Production Process Modeling / Simulation	New & Renewable Energy	Waste-to-Energy
Concept of Proposed Technology	This proposed technology develops a field-ready AI-driven prediction and optimization framework for waste plastic hydrothermal liquefaction (HTL). Instead of relying solely on new experiments, the project first constructs a comprehensive database from extensive literature and technical reports to build a foundational AI model. The Korean SME's specific hydrothermal process experiments and scale-up demonstration data are then used to validate, fine-tune, and refine the model (Transfer Learning). Crucially, to enhance on-site applicability, an AI-based sensing and image analysis module will be developed to identify plastic types during the pre-sorting stage. Using real-time feedstock data (from sensing) and operating conditions as inputs, the model predicts product distribution and quality. This integrated approach ensures technology meets industrial needs by providing practical solutions for feedstock variability and stable scale-up operation.			

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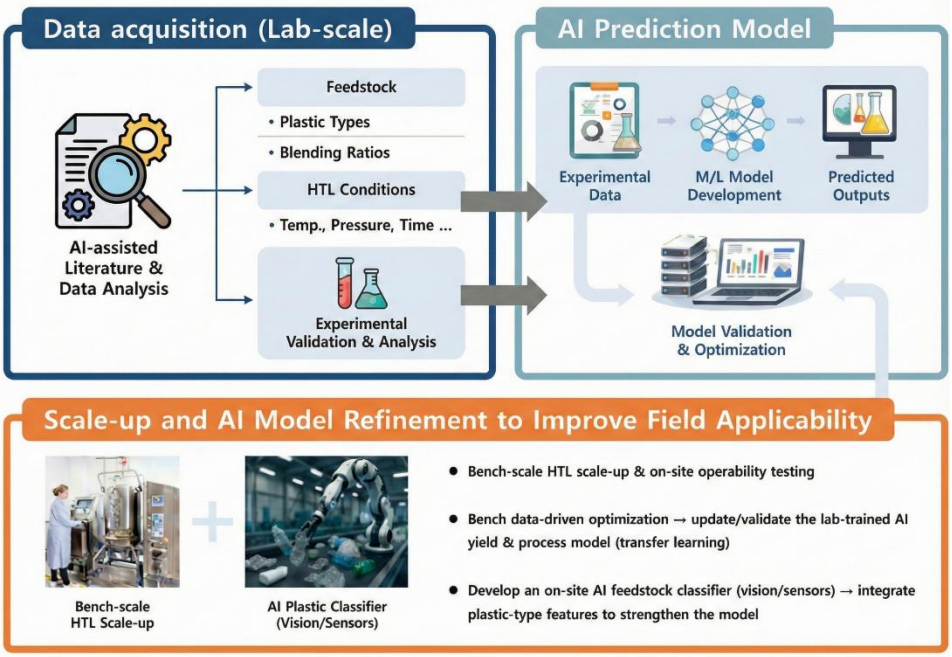

< Concept Diagram of the Field-Ready Hybrid AI Platform: Integrating Vision Sensing and Global Knowledge Mining ></td>
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<td>Rationale for Joint R&D Project</td>
<td>Mechanical recycling is currently the primary method for plastic waste management, but it faces critical limitations due to the high purity required for feedstock and the inevitable property degradation of mixed and contaminated post-consumer plastics [1,2]. While Hydrothermal Liquefaction (HTL) offers a promising chemical recycling alternative capable of handling wet and heterogeneous waste at 300–450 °C [3,5], its industrial deployment is hindered by two major challenges: (1) the inherent variability of mixed plastic waste entering the process is often unknown in real-time, leading to unstable operation, and (2) the discrepancy between lab-scale data and actual pilot-scale phenomena complicates reliable scale-up [4,6].

To overcome these hurdles and bridge the gap between laboratory research and industrial application, a "field-ready" AI approach is essential. Conventional AI modeling relies heavily on generating massive amounts of new experimental data, which is costly and time-consuming. Instead, this project proposes a novel framework that utilizes AI algorithms to automatically mine and aggregate global knowledge from extensive existing literature and technical reports, constructing a robust pre-trained foundation model without initial physical trials.

However, a literature-based model alone cannot address site-specific variables. Therefore, this international joint R&D integrates two critical field-oriented technologies: (1) an AI-based sensing and image analysis module to identify plastic types in real-time at the pre-sorting stage, and (2) targeted scale-up verification experiments.

Accordingly, this international joint R&D adopts an integrated approach that synergizes AI-driven data mining of global HTL literature with on-site AI vision sensing and targeted scale-up verification experiments. This strategy constructs a robust pre-trained foundation model and refines it through physical validation and fine-tuning (Transfer Learning), thereby delivering an optimized solution for real-world plant operations that effectively fuses global knowledge with high-fidelity field data.</td>
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Key Words	1. Waste plastics; 2. Hydrothermal liquefaction (HTL); 3. Crude oil yield prediction; 4. AI sensing 5. Transfer learning (field data-driven model refinement)

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (1)	**End of the Project**	Level (6)
R&D Objectives	**1. Global DB Construction & Targeted Validation** Construct a comprehensive HTL database by mining data from global literature and technical reports, followed by conducting targeted lab-scale and scale-up experiments to validate the pre-trained model and generate high-fidelity data for fine-tuning. **2. AI Sensing & Transfer Learning-based Prediction** Develop an AI-based vision sensing module to identify mixed plastic types in real-time, and validate predictive models (using Transfer Learning) that estimate product distribution (gas, solid, crude oil) and quality by integrating global knowledge with real-time sensing data. **3. Process Optimization & Scale-up Support** Develop an optimization and decision-support module that identifies robust operating windows to maximize crude oil yield and quality while minimizing variability caused by feedstock heterogeneity (detected by sensing), supporting reliable scale-up design from lab to pilot scale. **4. Integrated Field-Ready Workflow** Integrate the literature-based database, AI sensing module, predictive models, and optimization logic into a deployable 'Field-Ready' workflow to support AI-guided experiment planning (Active Learning), process design, and stable operation with continuous refinement.			
R&D Target Specifications	**1. Global HTL Database & Targeted Experimental Dataset** • Global DB Construction: Construct a database from literature/reports with ≥ 100 data points, covering diverse plastic types (PE, PP, PS, PET, etc.) and operating ranges. • Targeted Lab/Scale-up Data: Generate high-fidelity validation data (≥ 30 sets) focusing on "high uncertainty" regions identified by the pre-trained model. • Operating conditions: 250–350 °C, 30–90 min, water-to-plastic ratio 2–10. • Mass balance closure: 95–105% with CV ≤ 7% for crude oil yield in replicate runs. **2. AI Vision Sensing Performance** • Real-time Identification: Develop an AI vision module capable of classifying mixed plastic flakes (PE/PP/PS/PET/Others). • Accuracy Target: Achieve ≥85% Top-1 accuracy for PE/PP/PS/PET under controlled simulated conveyor conditions (fixed lighting, fixed camera distance, moderate belt speed,			

	single-layer/limited overlap). **3. Bench-scale HTL Operation & Scale-up Verification** • Stable Operation: Demonstrate stable bench-scale operation at ≥ 0.5–2.0 kg-feed/batch, proven by ≥ 5 consecutive batches (or ≥ 4 hours continuous equivalent). • Scale-up Factor Analysis: Quantify lab-to-bench differences (scale effects) and establish a bench operating window with documented SOPs, integrating real-time sensing data inputs. **4. Crude Oil Quality Characterization** • Composition Profiling: Determine crude oil composition using GC–MS (area% of aliphatics/aromatics) and HHV (MJ/kg). • Consistency: Ensure consistent identification criteria for top-N dominant compounds to support model training. **5. AI/ML Predictive Model Performance (Transfer Learning)** • Prediction Accuracy (Test Set): - Crude oil yield MAE ≤ 6 wt% ($R^2 \geq 0.80$) - HHV MAE ≤ 1.0 MJ/kg ($R^2 \geq 0.80$) • Transfer Learning Efficiency: Demonstrate that the pre-trained model (with fine-tuning) achieves the target accuracy with fewer experimental data points compared to a model trained from scratch. • External Validation: Quantify performance degradation on bench-scale data and report the causes (scale effects), targeting ≤40–50% MAE increase where feasible. **6. Optimization Module & Pilot Design Package** • Optimization Benefit: Demonstrate ≥ 10% relative improvement in crude oil yield or ≥ 20% reduction in variability across ≥ 3 mixed-feed scenarios. • Pilot Design Package: Deliver a complete PDP (PFD, P&ID, Mass/Energy Balance, Equipment List) with design basis derived from the AI-optimized bench results.
R&D Plan	**Overall Approach** The project adopts a "Literature-to-Field" strategy. It begins with (1) global data mining to build a pre-trained AI model and (2) developing an AI vision sensing module. This is followed by (3) targeted lab/bench-scale experiments to validate predictions and fine-tune the model (Transfer Learning). Finally, the integrated system is applied to (4) pilot-scale process design. The R&D loop is: (1) Literature DB, (2) Pre-trained Model, (3) AI Sensing Dev, (4) Targeted Validation, (5) Model Refinement, (6) Pilot Design. **<u>Year 1 (Months 1–12): DB Construction, AI Sensing Dev, and Targeted Validation (TRL 1 → 3)</u>** **[Key Tasks & Methodologies]** **1. Literature Data Mining & DB Construction (Months 1–8)**

- Construct a comprehensive HTL database (>100 points) by mining global literature/reports.
- Develop a pre-trained AI model and identify "knowledge gaps" to guide Korean SME's experiments.

2. AI Vision Sensing Module Development (Months 1–8)

- Develop a vision-based deep learning algorithm to classify plastic types (PE/PP/PS/PET) from flake images.
- Build a lab-scale sensing prototype for controlled, conveyor-simulated feedstock identification and logging.

3. Targeted Lab-scale Experiments (Months 9–12)

- Conduct lab-scale runs focused on "high uncertainty" conditions identified by the AI model.
- Validate the pre-trained model and quantify crude oil yield/quality (GC-MS/HHV) for fine-tuning.

[Milestones (Year 1)]

- M1: Global HTL Database constructed & AI Sensing algorithm design (Month 4)
- M2: Pre-trained AI Model developed & Sensing prototype built (Month 8)
- M3: Model validation via targeted lab experiments completed (Month 12)

Year 2 (Months 13–24): Bench Unit Set-up, Transfer Learning, and Optimization (TRL 3 → 4)

[Key Tasks & Methodologies]

1. Bench-scale Unit Build & Sensing Integration (Months 13–16)

- Commission bench unit (50L HTL Reactor) and install the AI vision sensing module at the feedstock inlet.
- Verify stable operation and real-time data logging from both the sensor and the reactor.

2. Bench-scale Campaigns & Scale-up Data (Months 17–20)

- Run bench-scale tests using mixed plastics identified by the AI sensor.
- Quantify lab-to-bench scale effects and generate field data for transfer learning.

3. Model Refinement (Transfer Learning) (Months 21–24)

- Fine-tune the pre-trained model using bench-scale & sensing data.
- Achieve target accuracy with minimized experimental data points (Efficiency validation).

4. Optimization Module (Months 21–24)

- Develop logic to recommend optimal set-points based on real-time sensing inputs to minimize yield variability.

[Milestones (Year 2)]

- M4: Bench unit commissioned + AI Sensing module integrated (Month 16)

	• M5: Bench dataset with sensing logs completed (Month 20) • M6: Bench-refined Model (Transfer Learning) & Optimization Module validated (Month 24) **Year 3 (Months 25–36): Pilot Design Package & Integrated Field-Ready Workflow (TRL 4 Achievement)** **[Key Tasks & Methodologies]** **1. Scale-Up Variables & Constraints Definition (Months 25–30)** • Formalize scale-up factors (heat transfer, mixing) and integrate them into the model as constraints. **2. Pilot-Scale Process Design Package (PDP) (Months 31–33)** • Deliver engineering outputs (PFD, P&ID, equipment sizing) based on the AI-optimized bench results. **3. Integrated "Field-Ready" Workflow (Months 34–36)** • Release the final software package: [AI Sensing] → [Real-time Prediction] → [Optimization]. • Provide user documentation for the "Auto-Design & Operation Guide". **4. Final Verification & TRL 4 Evidence (Months 34–36)** • Compile evidence of validated performance (Sensing accuracy + Yield prediction) and pilot design readiness. **[Milestones (Year 3)]** • M7: Scale-up variable framework completed (Month 30) • M8: Pilot Design Package (PDP) delivered (Month 33) • M9: Final Integrated Workflow released & TRL 4 verification report (Month 36)

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**The Research Institute** will lead the AI/ML-centered research activities, focusing on **global data mining, pre-trained model development, and transfer learning**, to provide a "field-ready" prediction and optimization framework. **1. Technical Contributions** • **Global Data Mining & Pre-training:** Lead the automated mining and aggregation of HTL data from global literature and technical reports to construct a comprehensive database and develop a pre-trained foundation model before physical experimentation. • **AI Vision Algorithm Development:** Develop deep learning algorithms for the AI vision sensing module to classify plastic types (PE/PP/PS/PET) from image data provided by the Company. • **Active Learning Strategy:** Implement uncertainty quantification to identify "knowledge gaps"

	and guide the Company's experimental design (Active Learning), minimizing trial-and-error. **2. Research Scope and Key Tasks** • **Data Ontology & Standardization:** Define the data structure and metadata requirements to unify diverse literature data and the Company's experimental data into a single coherent database. • **Transfer Learning & Refinement:** Perform transfer learning using the targeted lab/bench-scale data provided by the Company to fine-tune the pre-trained model for specific process conditions. • **Sensing-based Optimization:** Develop an optimization module that takes real-time sensing data as input to recommend robust operating windows that minimize yield variability caused by feedstock heterogeneity. **3. Resources to Be Provided** • Research manpower for AI/ML and Computer Vision development (PI supervision, graduate researchers). • Computing resources (HPC/Workstations) for training large-scale pre-trained models and vision algorithms. **4. Collaboration and Deliverables** • Deliverables: Pre-trained & Fine-tuned AI models, AI Vision Classification Algorithm, and "Experimental Design Guide" based on uncertainty analysis. • Feedback: Provide regular feedback on the Company's sensing data quality and experimental results to improve model accuracy.
Expected R&R of Korean SME Partner	**The Korean SME partner** will lead the **AI model implementation and targeted experimental verification**, generating high-fidelity field data to validate and fine-tune the AI models, and delivering the pilot-scale design package. **1. Technical Contributions and Development Tasks** • **AI Vision Model Implementation:** Implement and apply the plastic classification AI model to identify mixed plastic types (PE/PP/PS/PET) using image data acquired from the process, focusing on software integration rather than hardware fabrication. • **Targeted Validation Experiments:** Conduct lab-scale HTL experiments focused on high-uncertainty conditions identified by the Research Institute, generating high-quality dataset for model validation (target mass closure 95–105%). • **Crude Oil Characterization:** Perform HHV measurement and GC–MS profiling to provide ground-truth labels for the AI model. **2. Bench-Scale Scale-Up and Integration** • **Bench Unit & AI Integration:** Design and operate a 50L bench-scale HTL unit, integrating

	the AI classification model into the workflow to enable real-time feedstock identification and monitoring. • **Scale-up Data Generation:** Verify operability and assess scale effects by comparing lab- vs bench-scale yields, providing critical data for transfer learning. • **Field Constraints Identification:** Identify physical constraints (mixing limits, heat transfer lags) to refine the optimization logic. **3. Data Management and Collaboration** • **Data Provision:** Provide well-structured "Sensing-Process-Yield" paired datasets to the Research Institute. • **Model Validation:** Implement the recommended set-points from the AI optimization module in actual bench runs to validate performance improvements. **4. Pilot-Scale Design Package** • Process Engineering: Derive scale-up factors and operational requirements from the optimized bench results. • Design Package Delivery: Prepare the Pilot-Scale Process Design Package (PDP), including PFD, P&ID, mass & energy balance, and equipment sizing specifications. **5. Resources and Commitments** • Project manpower (process engineers, operators), lab/bench facilities (50L Reactor).

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Overcoming the "Valley of Death" in HTL Commercialization** Existing HTL technologies face a critical bottleneck: the **"Scale-up Discrepancy."** Lab-scale results often fail to replicate in continuous pilot plants due to the unpredictable variability of mixed waste feedstock and complex physical constraints. **This project delivers a paradigm shift: By integrating Global Knowledge (Literature Mining), Real-time "Eyes" (AI Sensing), and Physical Verification (Bench-scale)**, we provide a **"Risk-Minimized, Fast-Track Scale-up Solution"** that bridges the gap between fundamental research and industrial deployment. **1. Technological Outcomes: From "Black Box" to "Transparent Intelligence"** • **Solving the Feedstock Uncertainty Problem:** - Current Limit: Operators blindly feed mixed plastics, leading to unstable yields and reactor clogging. - Our Solution: Development of an "AI Vision Sensing Module" that identifies plastic types in real-time before they enter the reactor. - Outcome: Enabling feed-forward control strategies that adjust operating conditions dynamically, ensuring stable high-quality crude oil production even with heterogeneous waste. • **Global-Local Hybrid Intelligence (Transfer Learning):** - Current Limit: Building AI models from scratch requires years of expensive experiments.

- Our Solution: A "Literature-First, Experiment-Later" approach. We pre-train the model on global data (>100 points) and fine-tune it with high-fidelity Bench-scale data.
- Outcome: Achieving high prediction accuracy with 30-50% fewer physical experiments, proving a highly efficient R&D methodology.

- **Scale-up Ready Verification (The Power of Bench Implementation):**
  - Current Limit: Most academic AI studies rely solely on mg/g-scale lab data, ignoring heat transfer and mixing effects.
  - Our Solution: Validation through a 50L Bench-scale continuous unit, not just batch microtubes.
  - Outcome: A "Field-Proven" Pilot Design Package (PDP) that reflects real-world engineering constraints, significantly de-risking the subsequent TRL 5+ investment.

2. Commercialization Potential: Accelerating Time-to-Market

- **Drastic Reduction in R&D Costs & Time:**
  - By substituting trial-and-error experimentation with AI-guided Active Learning, we anticipate a >40% reduction in process development time and associated costs. This allows the SME to reach the Pilot Demonstration phase faster than competitors.
- **A "Turnkey" Solution for Chemical Recycling:**
  - The integrated workflow [Sensing → Prediction → Optimization] serves as a standalone software product or a licensing package for other chemical recycling plants, creating a new business model beyond oil production.
- **Investment Risk Mitigation:**
  - Providing a quantified "Operating Window" backed by scale-up verification gives investors and stakeholders confidence in the feasibility of the commercial plant, facilitating fundraising for the next-stage Pilot construction.

3. Broader Industrial and Economic Impact

- **Unlocking the Value of Low-Grade Waste:**
  - Enabling the processing of non-recyclable, mixed, and contaminated plastics that are currently incinerated or landfilled, directly contributing to the National Circular Economy Action Plan and carbon neutrality goals.
- **Strengthening Global Competitiveness of K-Eco Tech:**
  - Demonstrating a world-class reference where Korean Process Engineering meets Canadian AI Expertise. This success case will position the consortium as a leader in the global "Digital Transformation of Waste-to-Energy" sector.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization